

SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26044
- **Problem Statement Title** - Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** SIH26-A0H-T379
- **Team Name** (Registered on portal) – Tech Tonic



**ONGOING CHALLENGES**

Barriers in Academia – Industry Collaboration

RESUME

01 RESUME-BASED SKILLS

Skills are mostly **self-declared** through resumes, certificates and keyword lists, which **do not reflect actual practical ability**.

**02 NO PROOF OF COMPETENCY**

Student projects, GitHub work and other technical activities are **not systematically verified**, and individual contributions are often unclear.

**03 ACADEMIA – INDUSTRY GAP**

Industry requires specific, task-oriented skills, while academic learning and student profiles often **do not align** with real industry needs.

**04 INEFFICIENT MATCHING**

Existing portals rely on **keyword-based matching**, leading to a large number of applications with **uncertain skill fit**, causing both good candidates and opportunities to be missed.

**05 NO SKILL-GROWTH LOOP**

Students are **not clearly told** what skills they are missing or what actions to take. There is **limited conversion** of project outcomes into new, verified skill evidence.



Current systems focus on what students claim — not on what they can actually demonstrate.

**PROPOSED SOLUTION – SkillGraph**

From Student Work to Real Opportunities

Evidence.
Opportunities.
Growth.**01 AI-BASED EVIDENCE UNDERSTANDING**

Convert student work into structured skill information



Tech Used

LLM

NLP

RAG

02 AUTOMATED SKILL VERIFICATION

Prove skills through execution and testing



Tech Used

Static Analysis

Sandbox

Auto Testing

03 KNOWLEDGE GRAPH & INTELLIGENT MATCHING

Connect verified skills with industry opportunities



Tech Used

Knowledge Graph

Vector DB

Matching Engine

04 ADAPTIVE SKILL GAP & LEARNING

Identify gaps and create a path to grow



Tech Used

AI Assessment

Task Generation



More
Employable Students

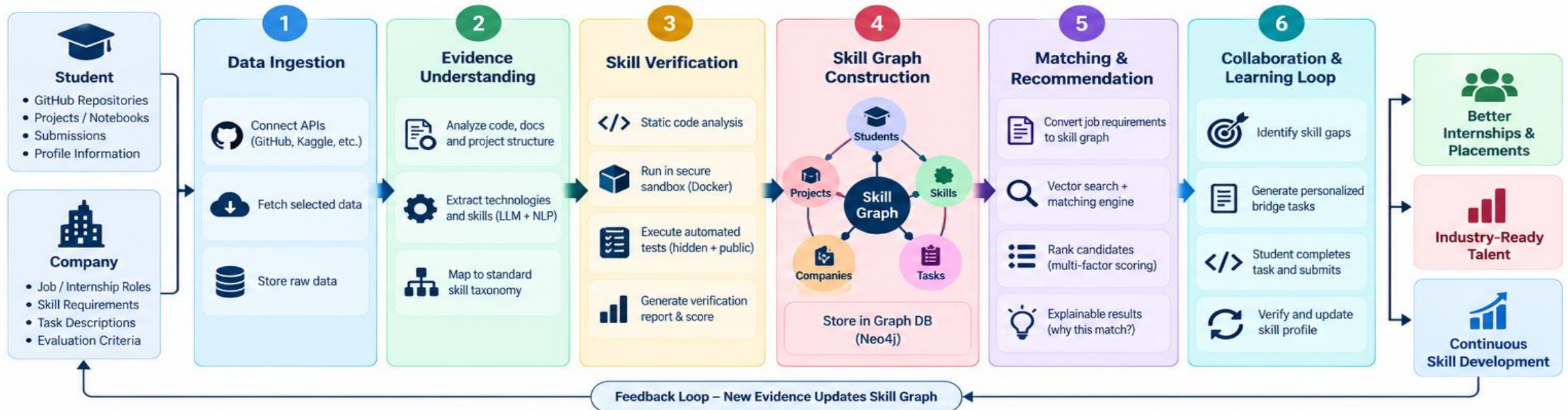


Stronger
Academia – Industry Collaboration



Work-Ready Talent
for Real-World Impact

BRIDGING TALENT AND OPPORTUNITY WITH REAL EVIDENCE



Technology Stack



Methodology & Process



WHAT WE COMMIT TO, AND WHY IT HOLDS

COMPONENT	OUR COMMITMENT	WHY IT IS FEASIBLE
 Data Ingestion	Connect to GitHub and other platforms or allow manual project uploads.	Uses existing APIs and open-source tools; minimal infrastructure.
 Skill Extraction & Understanding	Use LLM + NLP (with RAG) to extract technologies, skills and project context.	Proven LLM/NLP models and RAG frameworks; works on open-source data.
 Verification	Run automated tests in a secure sandbox to verify project functionality.	Containerization (Docker) enables safe and reproducible execution.
 Skill Graph	Build a graph connecting students, skills, projects, companies and tasks.	Graph databases (Neo4j/pg) are mature and widely used.
 Matching & Recommendation	Match verified skills with company requirements using vector search.	Vector databases and semantic search are scalable and effective.
 Deployment	Web platform accessible to students, colleges and companies.	Can be deployed on cloud/on-premise with modular architecture.

SCALE-OUT BOUNDARY — KEY DECISIONS


 Modular, Event-Driven Architecture


 Start with PostgreSQL + pgvector (scale to Neo4j/Qdrant)


 Containerized Verification Environment


 Versioned Artifacts & Continuous Model Improvement


POTENTIAL CHALLENGES AND RISKS

- Code Security** — Risk of malicious code or unsafe dependencies.
- AI Accuracy** — LLM may extract incorrect or incomplete information.
- Skill Verification** — Different project types need different evaluation methods.
- Contribution Tracking** — Difficult to assess individual contribution in team projects.
- Privacy & Fairness** — Unequal access to public data and privacy concerns.
- Scalability** — Large number of repositories and evaluation jobs.
- Industry Adoption** — Varying requirement formats and evaluation criteria.


STRATEGIES FOR OVERCOMING CHALLENGES

- Secure Execution** — Docker-based sandbox, no network access, resource limits and malware scanning.
- Reliable AI Insights** — Combine LLM with project evidence and use human review for uncertain cases.
- Flexible Verification** — Adapter-based tests for different project types (ML, Web, APIs, etc.).
- Contribution-Aware Assessment** — Analyze commit history and use challenge tasks when needed.
- Privacy & Fairness** — Consent-based data access and alternative assessment paths.
- Scalable Infrastructure** — Event-driven architecture with modular components and scalable databases.
- Industry Engagement** — Provide standard templates and collaborate with companies for clear requirements.

IMPACT ACROSS STAKEHOLDERS



STUDENTS

Learn • Build • Grow

1

Build verified skill profile
from real projects

2

Discover relevant
internships and projects

3

Identify skill gaps and
complete bridge tasks

4

Become industry-ready with
measurable outcomes

ACADEMIA

Guide • Bridge • Empower

1

Get data-driven view
of student skills

2

Identify curriculum–
industry skill gaps

3

Track student progress
through verified outcomes

4

Connect with industry mentors
and improve placement readiness

INDUSTRY

Find • Engage • Hire

1

Post real project requirements
and skill needs

2

Find candidates based on
demonstrated capabilities

3

Reduce irrelevant applications
and screening effort

4

Create a direct pathway
from project → internship → placement

KEY OUTCOMES — AND HOW THEY ARE MEASURED



STUDENT PLACEMENT READINESS

TARGET

≥ 50% more students with verified
skills profiles

HOW

Number of students with verified projects,
skill badges and completed assessments

INDUSTRY ENGAGEMENT

TARGET

≥ 40% reduction in irrelevant
applications

HOW

Compare number of shortlisted candidates
before and after using the platform

ACADEMIC ALIGNMENT

TARGET

≥ 30% better visibility into
curriculum–industry skill gaps

HOW

Track usage of skill gap reports and
curriculum recommendations

TIME TO HIRE / MATCH

TARGET

≥ 50% faster candidate–opportunity
matching

HOW

Measure average time from requirement
posting to shortlist / offer

LONG-TERM IMPACT

Stronger academia–industry
collaboration

Evidence-driven skill mapping and hiring



A continuous pipeline of work-ready talent



Higher employability and innovation ecosystem

SOURCES — EACH ONE SUPPORTS A DESIGN DECISION

SOURCE	WHAT IT ESTABLISHES	WHERE IT LANDS IN OUR DESIGN
Neo4j Graph Database Documentation neo4j.com/docs	Establishes graph-based data modeling and querying for connected data.	Used for the SkillGraph layer connecting students, skills, projects, companies and tasks.
NIST AI Risk Management Framework nvlpubs.nist.gov/nistpubs/ai	Establishes risk-aware and trustworthy AI practices.	Guides human review, provenance, confidence-aware AI outputs and overall governance.
Docker Engine Security Documentation docs.docker.com/security	Establishes container isolation and security controls for running untrusted code.	Used for secure sandbox execution during project verification.
W3C Verifiable Credentials Data Model www.w3.org/TR/vc-data-model	Establishes portable, machine-verifiable credentials.	Supports the Evidence Passport with verifiable skill evidence for students.
OWASP Top 10 Web Application Security Risks owasp.org/www-project-top-ten	Establishes common web application security risks and mitigations.	Informs secure API design, authentication, authorization and overall platform security.
Qdrant Vector Database Documentation qdrant.tech/documentation	Establishes vector similarity search infrastructure.	Supports semantic matching between company requirements and student evidence.

TECHNICAL FOUNDATION

Technology choices are grounded in the project architecture and implementation plan.

FastAPI Backend APIs for platform services and integrations	PostgreSQL Transactional data for users, projects and workflows	Temporal Long-running, auditable workflows for asynchronous processing	Kubernetes Jobs Scalable, isolated evaluation environment for project verification
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